



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 7: Consumer Theory V

Patricia Galilea A.

ICS2523 Microeconomics

Lecture 7

Last lecture we

- Learnt important identities
- Started analysing **changes in Income and Prices**

In today's lecture we will

- Learn the **Slutsky equation**
- Review Slutsky equation and its derivation
- Do **exercises** to practice these new concepts

Questions from last lecture

Concepts that need review:

- Substitution effect: how a change in the price of a good affects demand compared to others.
- Income effect: how a change in demand for a good is affected by a change in real disposable income

These definitions are incorrect because those effects are not total; they are used to explain a price change.

Changes in good's price

Consumer Theory

8. Changes in a good's Price (Review)

Since a **change in a good's price** changes the intercept of the budget constraint AND its slope, this is more complex.

Two effects come into play:

1. **Substitution Effect**
2. **Income Effect**

8. Changes in a good's Price

(Review)

If the price of x decreases

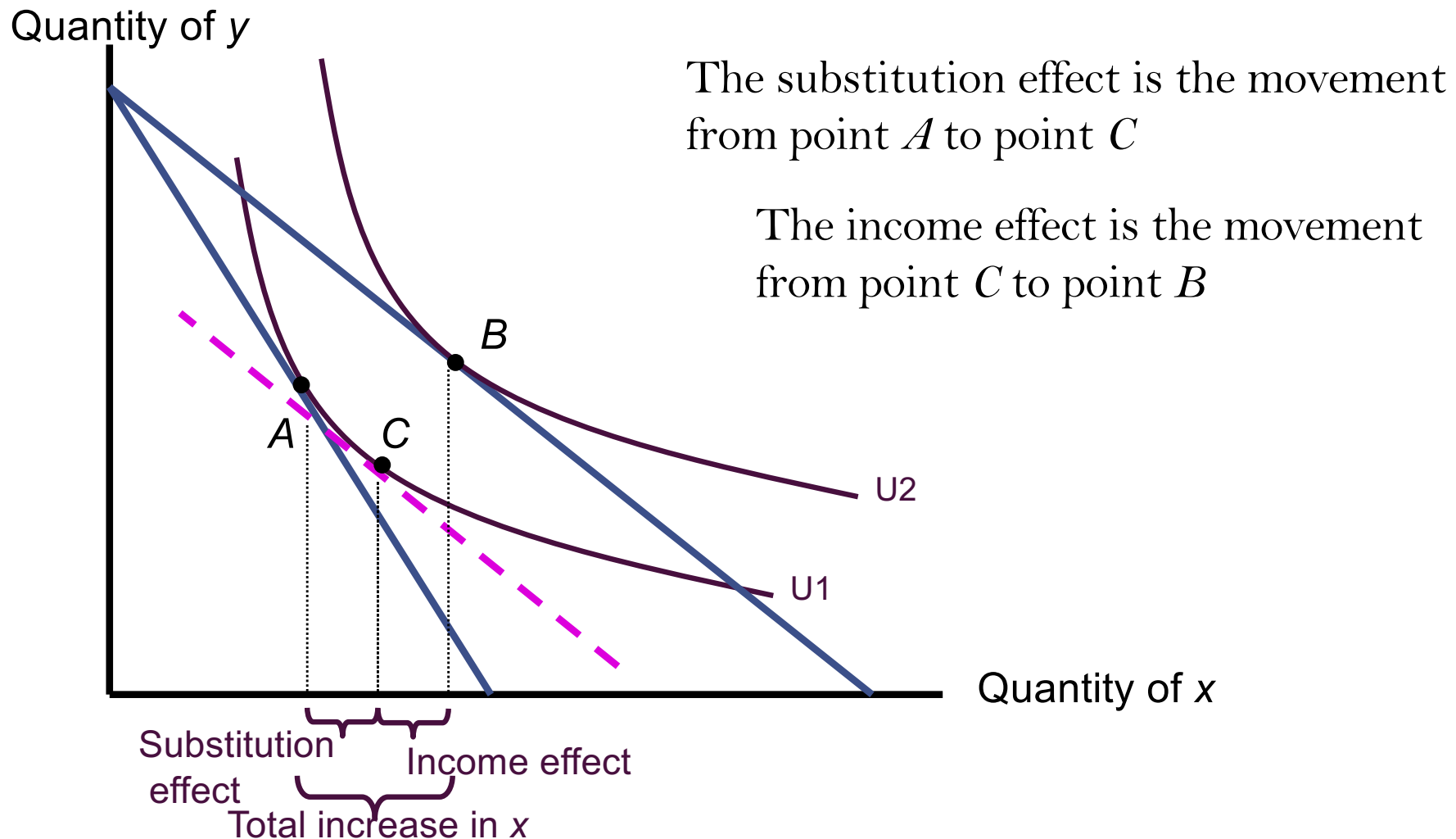
Substitution Effect: consumer should tend to buy more of that good and less from the others since they are now relatively more expensive $\rightarrow$ change on consumption of x_i associated with a change of its price at the same level of utility (i.e. same indifference curve, but now consumption patterns are allocated to equate MRS to the new price ratio)

Income Effect: since now one of the goods is cheaper, the consumer has more purchasing power $\rightarrow$ change on x_i consumption due to a change in his purchasing power (i.e., the individual cannot stay on the initial indifference curve and must move to a new one).

8. Changes in a good's Price

(Review)

P_x falls



8. Price Changes for Normal Goods (Review)

If a good is **normal**, substitution and income effects reinforce one another

- when price falls, both effects lead to a rise in quantity demanded
- when price rises, both effects lead to a drop in quantity demanded

8. Price Changes for Inferior Goods (Review)

If a **good is inferior**, substitution and income effects move in opposite directions

The combined effect is indeterminate

- when price rises, the substitution effect leads to a drop in quantity demanded, but the income effect is opposite
- when price falls, the substitution effect leads to a rise in quantity demanded, but the income effect is opposite

8. Giffen's Paradox

(Review)

If the income effect of a price change is strong enough, there could be a positive relationship between price and quantity demanded

- an increase in price leads to a drop in real income
- since the good is inferior, a drop in income causes quantity demanded to rise

8. Changes in a Good's Price

(Review)

Substitution Effect:

- If $P_x \uparrow \rightarrow \downarrow x$
- If $P_x \downarrow \rightarrow \uparrow x$

Income Effect: Depends on the type of good

- If $P_x \uparrow$ ($\downarrow$ purchasing power)
 - If it is a normal good $\rightarrow \downarrow x$
 - If it is an inferior good $\rightarrow \uparrow x$
- If $P_x \downarrow$ ($\uparrow$ purchasing power)
 - If it is a normal good $\rightarrow \uparrow x$
 - If it is an inferior good $\rightarrow \downarrow x$
- Giffen goods are a subset of inferior goods, where the Income Effect is stronger than substitution effect.

Activity 1

If the price of x decreases:

1. What happens to the consumption of x if it is a normal good?
 - a. Rises
 - b. Falls
 - c. Nothing
 - d. It depends
2. And what happens to the consumption of x if it is an inferior good?
 - a. Rises
 - b. Falls
 - c. Nothing
 - d. It depends



Wooclap: **OIQJXX**

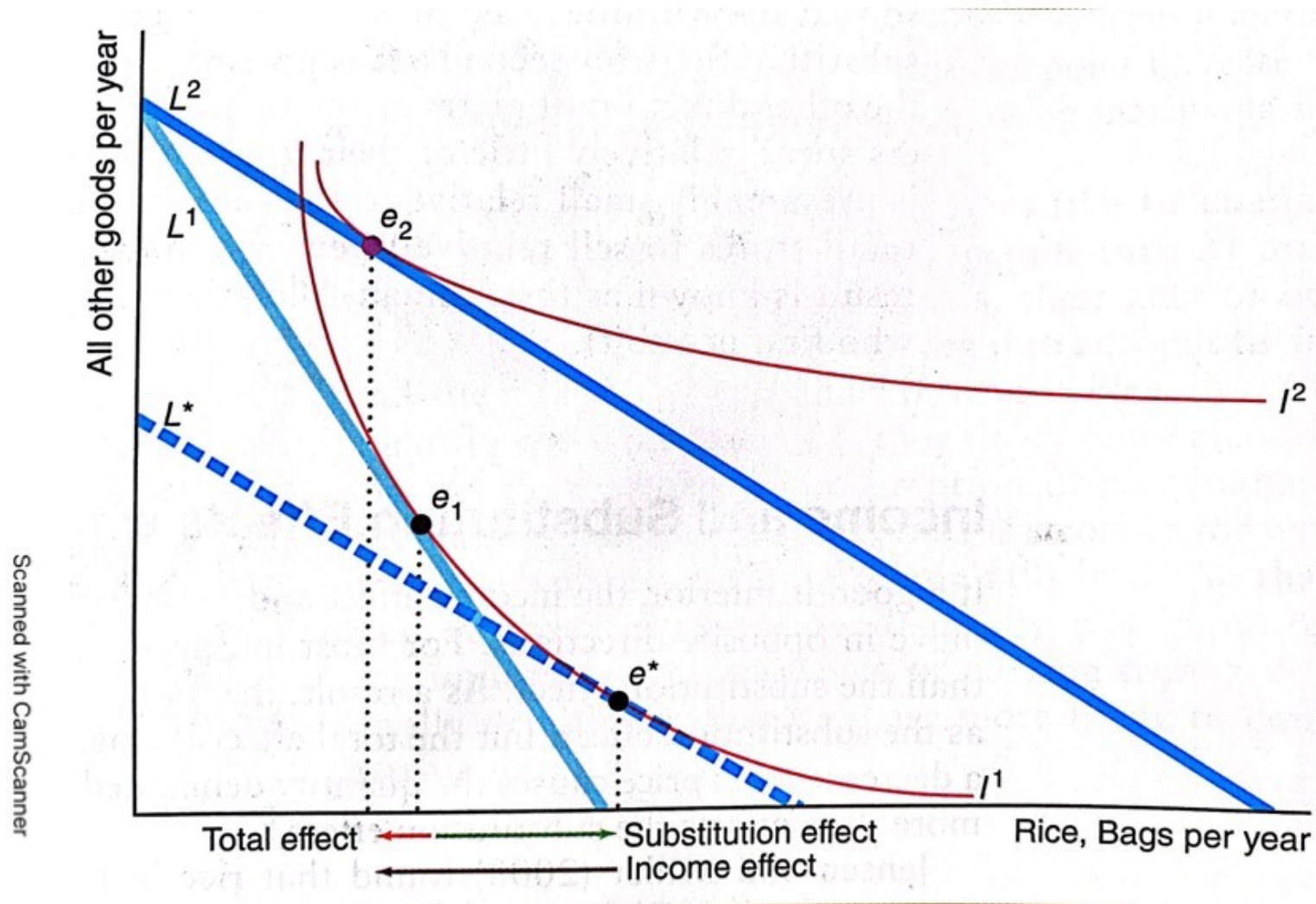
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Activity 2

Analyse graphically what happens with x consumption if there is a decrease of P_x and x is Giffen good.



Giffen example (inspired in rice)



Slutsky Equation

Consumer Theory

9. Slutsky equation

The **Slutsky Equation** is a mathematical development of response to price changes

We wish to calculate $\frac{\partial x_i}{\partial P_j}$

Slutsky equation:

$$\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial P_j} = \frac{\partial h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))}{\partial P_j} - \frac{\partial x_i(P_1, \dots, P_n, I)}{\partial I} x_j(P_1, \dots, P_n, I)$$

9. Slutsky equation

Proof:

Let x^* maximize utility at (P^*, m^*) and let $U^* = U(x^*)$.
From identity (6) from Lecture 5

$$h_i(P_1, \dots, P_n, U^*) \equiv x_i(P_1, \dots, P_n, E(P_1, \dots, P_n, U^*))$$

If we differentiate this with respect to P_j and evaluate the derivative at P^*

$$\frac{\partial h_i(P_1^*, \dots, P_n^*, U)}{\partial P_j} = \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial P_j} + \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial I} \boxed{\frac{\partial E(P_1^*, \dots, P_n^*, U^*)}{\partial P_j}}$$

Consumer Theory

$$\frac{\partial h_i(P_1^*, \dots, P_n^*, U^*)}{\partial P_j} = \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial P_j} + \frac{\partial x_i(P_1^*, \dots, P_n^*, E(P_1^*, \dots, P_n^*, U^*))}{\partial E(P_1^*, \dots, P_n^*, U^*)} \frac{\partial E(P_1^*, \dots, P_n^*, U^*)}{\partial P_j}$$

According to Shephard's Lemma:

$$\frac{\partial E(P_1^*, \dots, P_n^*, U^*)}{\partial P_j} = h_j(P_1^*, \dots, P_n^*, U^*)$$

$$\frac{\partial h_i(P_1^*, \dots, P_n^*, U^*)}{\partial P_j} = \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial P_j} + \frac{\partial x_i(P_1^*, \dots, P_n^*, E(P_1^*, \dots, P_n^*, U^*))}{\partial E(P_1^*, \dots, P_n^*, U^*)} h_j(P_1^*, \dots, P_n^*, U^*)$$

Since $I^* = E(P_1^*, \dots, P_n^*, U^*)$

$$\frac{\partial h_i(P_1^*, \dots, P_n^*, U^*)}{\partial P_j} = \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial P_j} + \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial I^*} h_j(P_1^*, \dots, P_n^*, U^*)$$

$$\frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial P_j} = \frac{\partial h_i(P_1^*, \dots, P_n^*, U^*)}{\partial P_j} - h_j(P_1^*, \dots, P_n^*, U^*) \frac{\partial x_i(P_1^*, \dots, P_n^*, I^*)}{\partial I^*}$$

Consumer Theory

$$\frac{\partial x_i (P_1^*, \dots, P_n^*, I^*)}{\partial P_j} = \frac{\partial h_i(P_1^*, \dots, P_n^*, U^*)}{\partial P_j} - h_j(P_1^*, \dots, P_n^*, U^*) \frac{\partial x_i (P_1^*, \dots, P_n^*, I^*)}{\partial I^*}$$

Using identity 5 from Lecture 5:

$$x_i(P_1, \dots, P_n, I) \equiv h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))$$

$$\frac{\partial x_i (P_1^*, \dots, P_n^*, I^*)}{\partial P_j} = \frac{\partial h_i(P_1^*, \dots, P_n^*, U^*)}{\partial P_j} - x_j(P_1^*, \dots, P_n^*, I^*) \frac{\partial x_i (P_1^*, \dots, P_n^*, I^*)}{\partial I^*}$$

Slutsky equation:

$$\frac{\partial x_i (P_1^*, \dots, P_n^*, I^*)}{\partial P_j} = \underbrace{\frac{\partial h_i(P_1^*, \dots, P_n^*, U^*)}{\partial P_j}}_{\text{Substitution Effect}} - \underbrace{x_j(P_1^*, \dots, P_n^*, I^*) \frac{\partial x_i (P_1^*, \dots, P_n^*, I^*)}{\partial I^*}}_{\text{Income Effect}}$$

Notation

How to write the quantity demanded?

- The quantity demanded is written as x or q depending on the text.
- For a two -goods case, it is common to use x and y .
- For a n-goods case, for referring to the demand of good i is more common to refer to x_i , where $i = 1 \text{ to } n$

Notation

Marshallian demand

- Since it is the most widely used demand, it is often called x and y . Also, it could be referred to as x_m

Hicksian demand

- It is often referred to as x_h ; in other books, it is called h . In our reference book (Nicholson), it is denoted as the following expression when explaining the substitution effect in the Slutsky equation:

$$\left. \frac{\partial x}{\partial P_x} \right|_{U=\text{constant}}$$

Here, the author is referring to the hicksian demand because he is keeping constant the level of utility.

Gross & Net Complements and Substitutes

- **Gross Complements:** Total effect on good x_i of increase of P_j is negative

$$\frac{\partial x_i}{\partial P_j} < 0$$

- **Net Complements:** looking only at the substitution effect (keeping constant the utility level), the impact on good x_i of increase of P_j is negative

$$\left. \frac{\partial h_i}{\partial P_j} \right|_{U=\bar{U}} < 0$$

- **Gross substitutes:** Total effect on good x_i of increase of P_j is positive

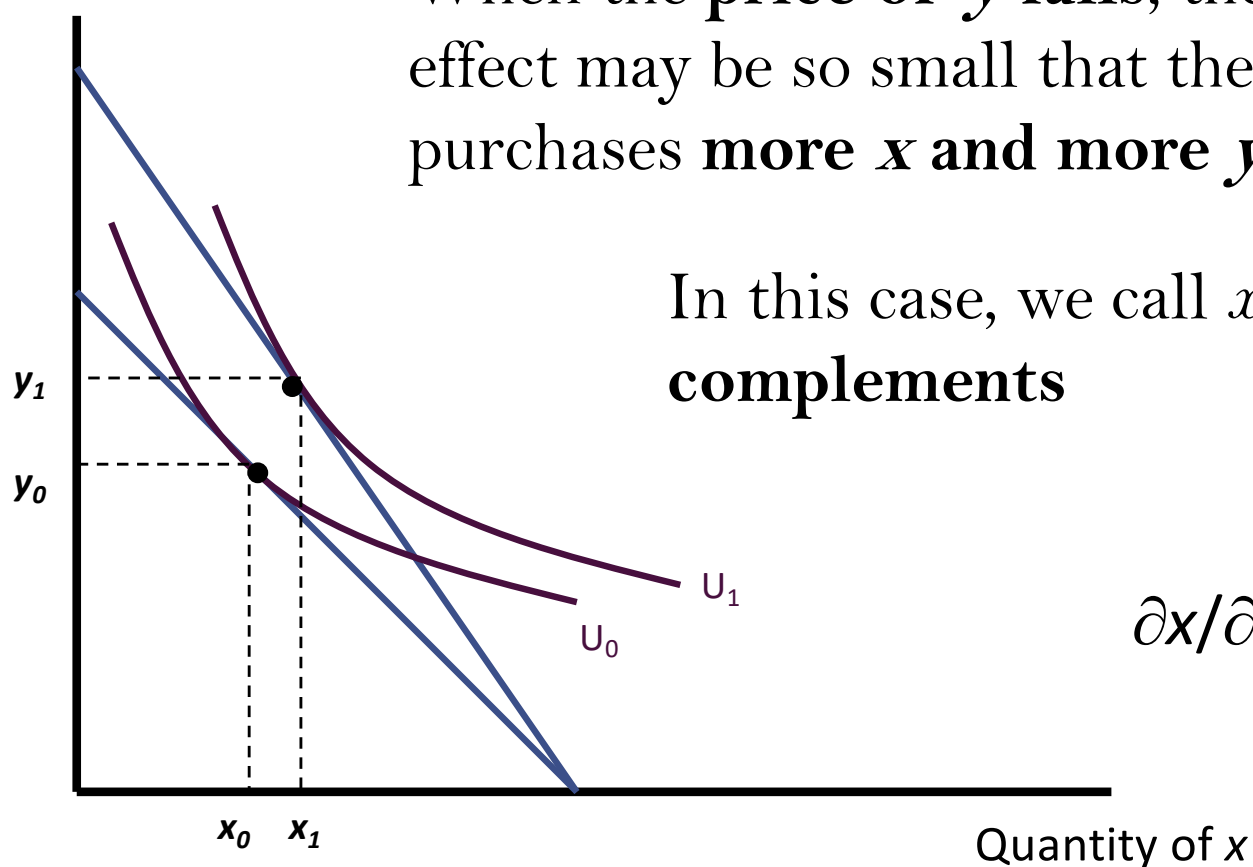
$$\frac{\partial x_i}{\partial P_j} > 0$$

- **Net substitutes:** looking only at the substitution effect (keeping constant the utility level), the effect on good x_i of increase of P_j is positive

$$\left. \frac{\partial h_i}{\partial P_j} \right|_{U=\bar{U}} > 0$$

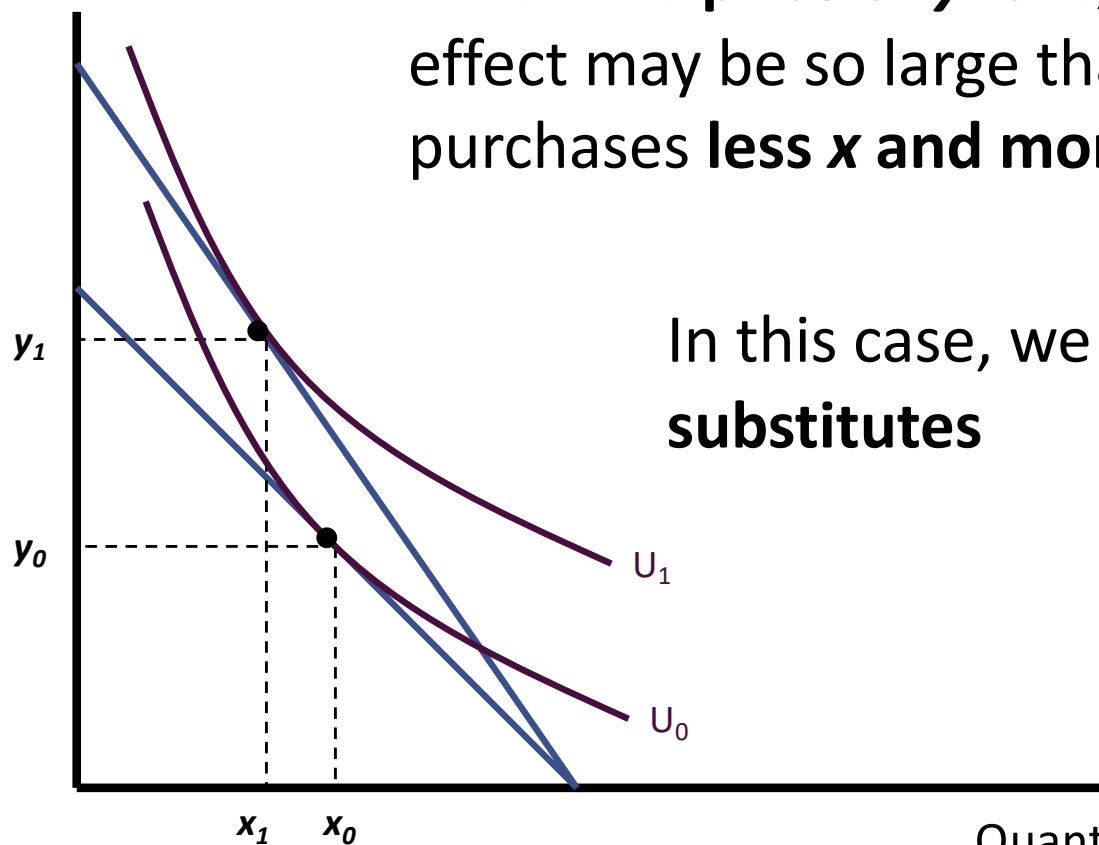
Gross Complements

Quantity of y



Gross Substitutes

Quantity of y



Asymmetry of the Gross Definitions

One undesirable characteristic of the gross definitions of substitutes and complements is that they are not symmetric

It is possible for x_1 to be a substitute for x_2 and at the same time for x_2 to be a complement of x_1

Asymmetry of the Gross Definitions

Suppose that the utility function for two goods is given by

$$U(x,y) = \ln x + y$$

Setting up the Lagrangian

$$\mathbf{L} = \ln x + y + \lambda(I - p_x x - p_y y)$$

Asymmetry of the Gross Definitions

gives us the following first-order conditions:

$$\partial \mathbf{L} / \partial x = 1/x - \lambda p_x = 0$$

$$\partial \mathbf{L} / \partial y = 1 - \lambda p_y = 0$$

$$\partial \mathbf{L} / \partial \lambda = I - p_x x - p_y y = 0$$

Manipulating the first two equations, we get

$$p_x x = p_y y$$

Asymmetry of the Gross Definitions

Inserting this into the budget constraint, we can find the Marshallian demand for y

$$p_x y = I - p_y$$

an increase in p_y causes a decline in spending on y

- Since p_x and I are unchanged, spending on x must rise ($\Rightarrow$ x and y are gross substitutes)
- But spending on y is independent of p_x ($\Rightarrow$ x and y are independent of one another)

Net Substitutes and Complements

The concepts of net substitutes and complements focus solely on substitution effects:

- Two goods are **net substitutes** if

$$\left. \frac{\partial h_i}{\partial p_j} \right|_{U=\text{constant}} > 0$$

- Two goods are **net complements** if

$$\left. \frac{\partial h_i}{\partial p_j} \right|_{U=\text{constant}} < 0$$

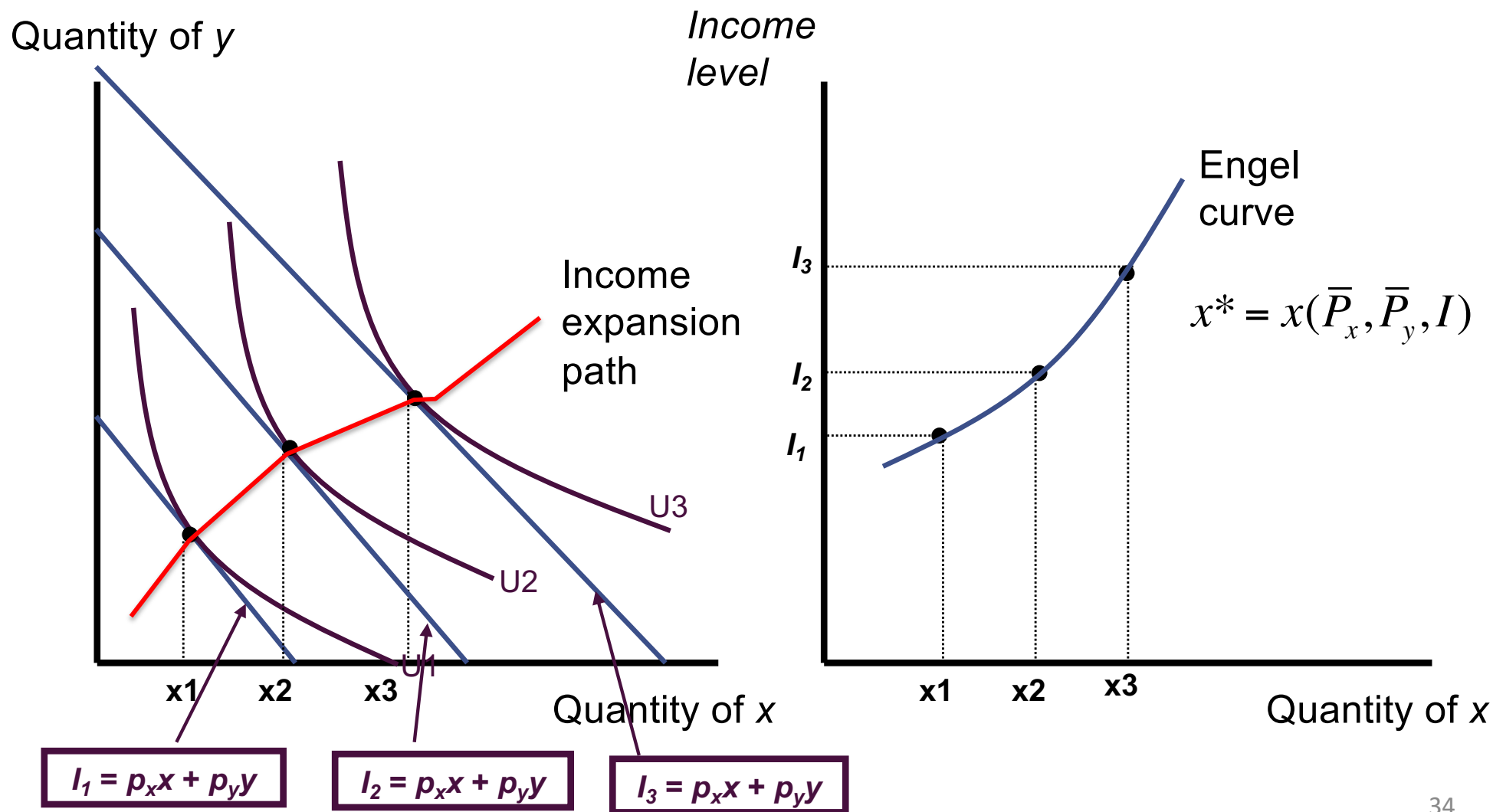
Net Substitutes and Complements

- This definition looks only at the shape of the indifference curve
- This definition is unambiguous because the definitions are perfectly symmetric

$$\left. \frac{\partial h_i}{\partial p_j} \right|_{U=\text{constant}} = \left. \frac{\partial h_j}{\partial p_i} \right|_{U=\text{constant}}$$

Income expansion path versus Engel curves

Let's see what happens when Income varies (ceteris paribus):



Example: Perfect Complements

If the utility function of an individual is $U(x, y) = \min(\alpha x, \beta y)$ find his:

- Find his Marshallian demand function. Verify that it is homogenous of degree zero in prices and income.
- Find his Indirect utility function. Verify that it is homogenous of degree zero in prices and income.
- Find his Hicksian demand function. Verify that it is homogenous of degree zero in prices.
- Find his Expenditure function.

Finally,

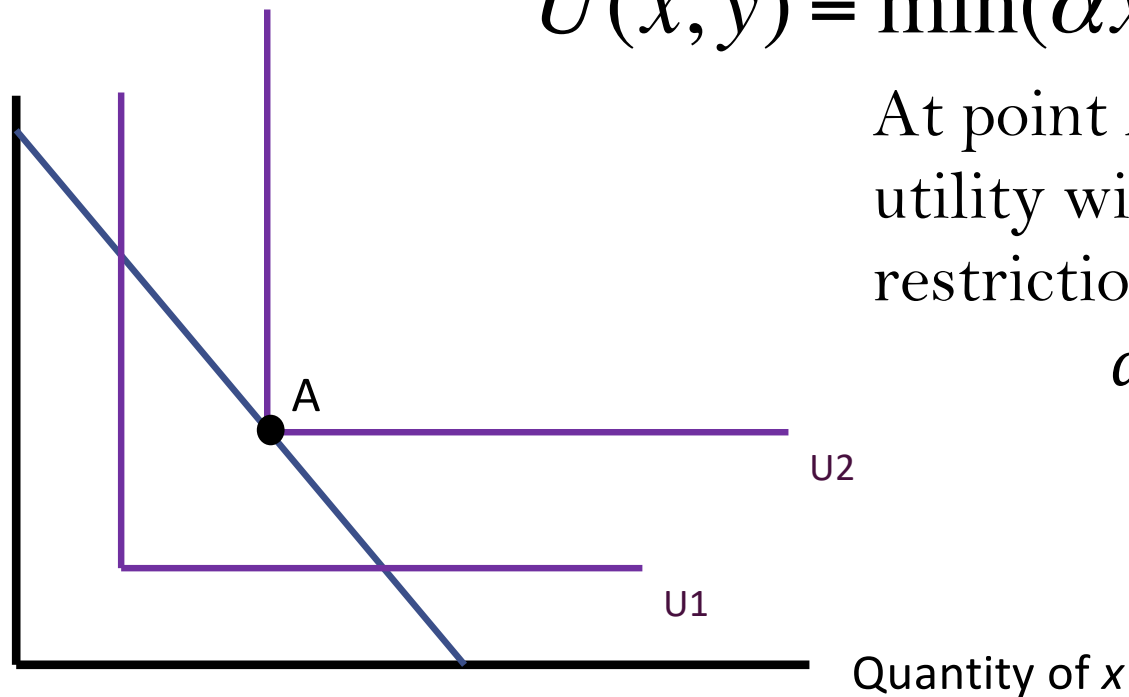
- Verify Roy's identity and Shephard's lemma.
- Verify Slutsky's equation.

Example L7: solution

If we graph the indifference curves representing perfect complements, we will see that its MRS will differ from the budget restriction's slope $\rightarrow$ not an interior solution

$$U(x, y) = \min(\alpha x, \beta y)$$

Quantity of y



At point A we maximize utility within the budget restriction.

$$\alpha x = \beta y$$

Example L7: solution

Since $\alpha x = \beta y$, $x = \frac{\beta y}{\alpha}$

Replacing this expression in the budget constraint:

$$\begin{aligned} P_x x + P_y y &= I \\ P_x \frac{\beta y}{\alpha} + P_y y &= I \\ y_m^* &= I \frac{\alpha}{\beta P_x + \alpha P_y} \end{aligned}$$

Similarly,

$$x_m^* = I \frac{\beta}{\beta P_x + \alpha P_y}$$

Example L7: solution

Are these Marshallian demands homogenous of degree zero in prices and income?

$$x_m^*(aP_x, aP_y, aI) = a^0 x_m^*(P_x, P_y, I)$$

$$x_m^*(aP_x, aP_y, aI) = (aI) \frac{\beta}{\beta aP_x + \alpha aP_y} = x_m^*(P_x, P_y, I)$$

Example L7: solution

The indirect utility function will be:

$$V(P_x, P_y, I) = \min(\alpha x_m^*, \beta y_m^*) \\ = \min(\alpha I \frac{\beta}{\beta P_x + \alpha P_y}, I \beta \frac{\alpha}{\beta P_x + \alpha P_y}) = I \frac{\alpha \beta}{\beta P_x + \alpha P_y}$$

Is this indirect utility function homogenous of degree zero in prices and income?

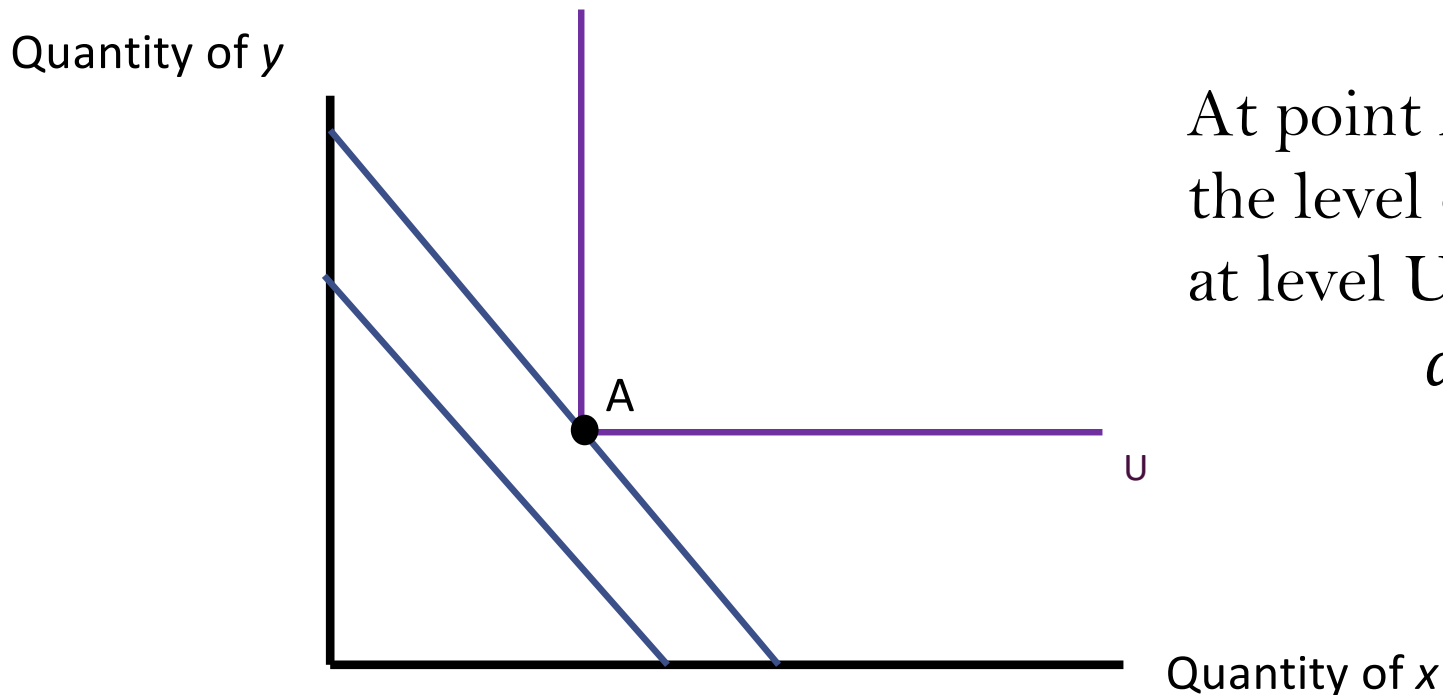
$$V(aP_x, aP_y, aI) = a^0 V(P_x, P_y, I)$$

$$V(aP_x, aP_y, aI) = (aI) \frac{\alpha \beta}{\beta aP_x + \alpha aP_y} = a^0 I \frac{\alpha \beta}{\beta P_x + \alpha P_y}$$

Example L7: solution

The dual problem is solved graphically, too. Now the level of utility is fixed, and we move from one level of expenditure to another.

$$U(x, y) = \min(\alpha x, \beta y)$$



At point A we minimize the level of expenditure at level U of utility.

$$\alpha x = \beta y$$

Example L7: solution

Since $\alpha x = \beta y$, $x = \frac{\beta y}{\alpha}$

Replacing this expression in the level of utility:

$$U = \min \left(\alpha \frac{\beta y}{\alpha}, \beta y \right) = \beta y$$

$$y_h^* = \frac{U}{\beta}$$

Similarly,

$$x_h^* = \frac{U}{\alpha}$$

Example L7: solution

Are these Hicksian demands homogenous of degree zero in prices?

$$x_h^*(aP_x, aP_y, U) = a^0 x_h^*(P_x, P_y, U)$$

$$x_h^*(aP_x, aP_y, U) = \frac{U}{\alpha} = a^0 x_m^*(P_x, P_y, I)$$

The expenditure function is:

$$E(P_x, P_y, U) = P_x x_h^* + P_y y_h^* = \left(\frac{P_x}{\alpha} + \frac{P_y}{\beta} \right) U$$

Example L7: solution

$$\text{Roy's identity: } x_m^* = - \frac{\frac{\partial V(P_x, P_y, I)}{\partial P_x}}{\frac{\partial V(P_x, P_y, I)}{\partial I}}$$

$$\begin{aligned} \frac{\partial V(P_x, P_y, I)}{\partial P_x} &= \frac{\partial (I \frac{\alpha\beta}{\beta P_x + \alpha P_y})}{\partial P_x} = I\alpha\beta(-1) \frac{1}{(\beta P_x + \alpha P_y)^2} \beta \\ \frac{\partial V(P_x, P_y, I)}{\partial I} &= \frac{\partial (I \frac{\alpha\beta}{\beta P_x + \alpha P_y})}{\partial I} = \frac{\alpha\beta}{\beta P_x + \alpha P_y} \end{aligned}$$

Thus,

$$x_m^* = - \frac{\frac{\partial V(P_x, P_y, I)}{\partial P_x}}{\frac{\partial V(P_x, P_y, I)}{\partial I}} = - \frac{I\alpha\beta(-1) \frac{1}{(\beta P_x + \alpha P_y)^2} \beta}{\frac{\alpha\beta}{\beta P_x + \alpha P_y}}$$

Example L7: solution

$$\text{Roy's identity: } x_m^* = - \frac{\frac{\partial V(P_x, P_y, I)}{\partial P_x}}{\frac{\partial V(P_x, P_y, I)}{\partial I}}$$

$$x_m^* = - \frac{I\alpha\beta(-1) \frac{1}{(\beta P_x + \alpha P_y)^2} \beta}{\frac{\alpha\beta}{\beta P_x + \alpha P_y}} = I \frac{\beta}{\beta P_x + \alpha P_y}$$

Example L7: solution

Shephard's Lemma: $x_h^* = \frac{\partial E(P_x, P_y, U)}{\partial P_x}$

$$E(P_x, P_y, U) = P_x x_h^* + P_y y_h^* = \left(\frac{P_x}{\alpha} + \frac{P_y}{\beta} \right) U$$

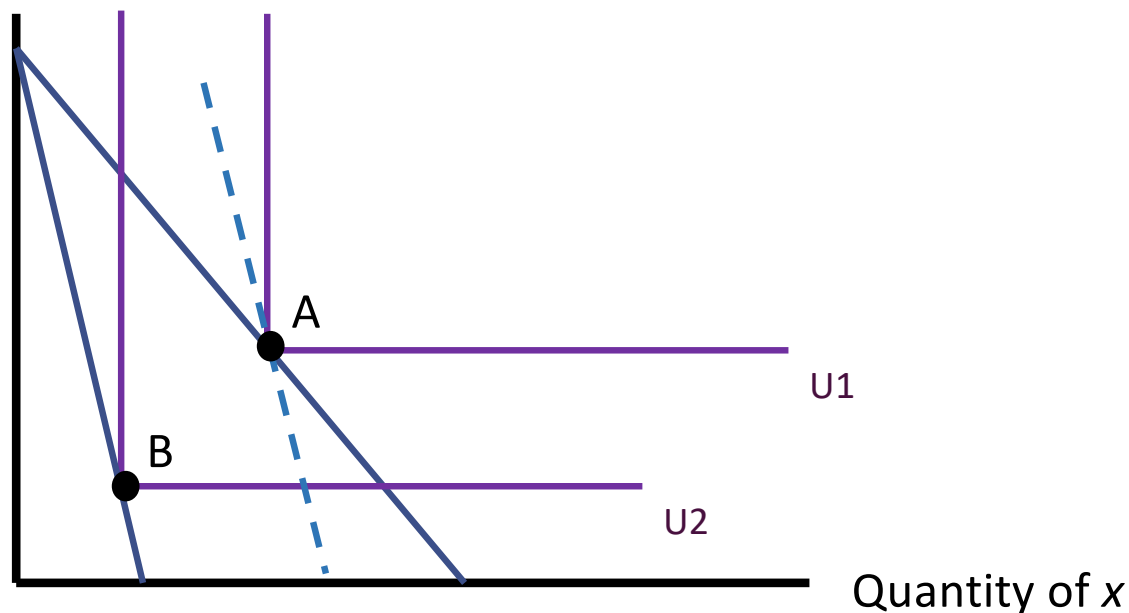
$$\frac{\partial E(P_x, P_y, U)}{\partial P_x} = \frac{U}{\alpha} = x_h^*$$

Example L7: solution

Slutsky Equation:

$$\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial P_j} = \underbrace{\frac{\partial h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))}{\partial P_j}}_{\text{SUBSTITUTION EFFECT}} - \underbrace{\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial I} x_j(P_1, \dots, P_n, I)}_{\text{INCOME EFFECT}}$$

Quantity of y



Example L7: solution

$$\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial P_j} = \frac{\partial h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))}{\partial P_j} - \frac{\partial x_i(P_1, \dots, P_n, I)}{\partial I} x_j(P_1, \dots, P_n, I)$$

$$\frac{\partial x_h^*}{\partial P_x} = \frac{\partial \left(\frac{U}{\alpha} \right)}{\partial P_x} = 0$$

**SUBSTITUTION
EFFECT**

$$\frac{\partial x_m^*}{\partial I} = \frac{\partial \left(I \frac{\beta}{\beta P_x + \alpha P_y} \right)}{\partial I} = \frac{\beta}{\beta P_x + \alpha P_y}$$

$$\frac{\partial x_m^*}{\partial P_x} = \frac{\partial \left(I \frac{\beta}{\beta P_x + \alpha P_y} \right)}{\partial P_x} = -I \frac{\beta}{(\beta P_x + \alpha P_y)^2} \beta$$

Example L7: solution

$$\frac{\partial x_i(P_1, \dots, P_n, I)}{\partial P_j} = \frac{\partial h_i(P_1, \dots, P_n, V(P_1, \dots, P_n, I))}{\partial P_j} - \frac{\partial x_i(P_1, \dots, P_n, I)}{\partial I} x_j(P_1, \dots, P_n, I)$$

$$-I \frac{\beta^2}{(\beta P_x + \alpha P_y)^2} = 0 - \frac{\beta}{\beta P_x + \alpha P_y} I \frac{\beta}{\beta P_x + \alpha P_y}$$

$$-I \frac{\beta^2}{(\beta P_x + \alpha P_y)^2} = 0 - I \frac{\beta^2}{(\beta P_x + \alpha P_y)^2}$$

Homework: Cobb-Douglas

If the utility function of an individual is $U(x, y) = x^{0.4}y^{0.6}$ find his/hers:

- Find his Marshallian demand function. Verify that it is homogenous of degree zero in prices and income.
- Find his Indirect utility function. Verify that it is homogenous of degree zero in prices and income.
- Find his Hicksian demand function. Verify that it is homogenous of degree zero in prices.
- Find his Expenditure function.

Finally,

- Verify that Roy's identity and Shephard's lemma.
- Verify Slutsky's equation.

Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.

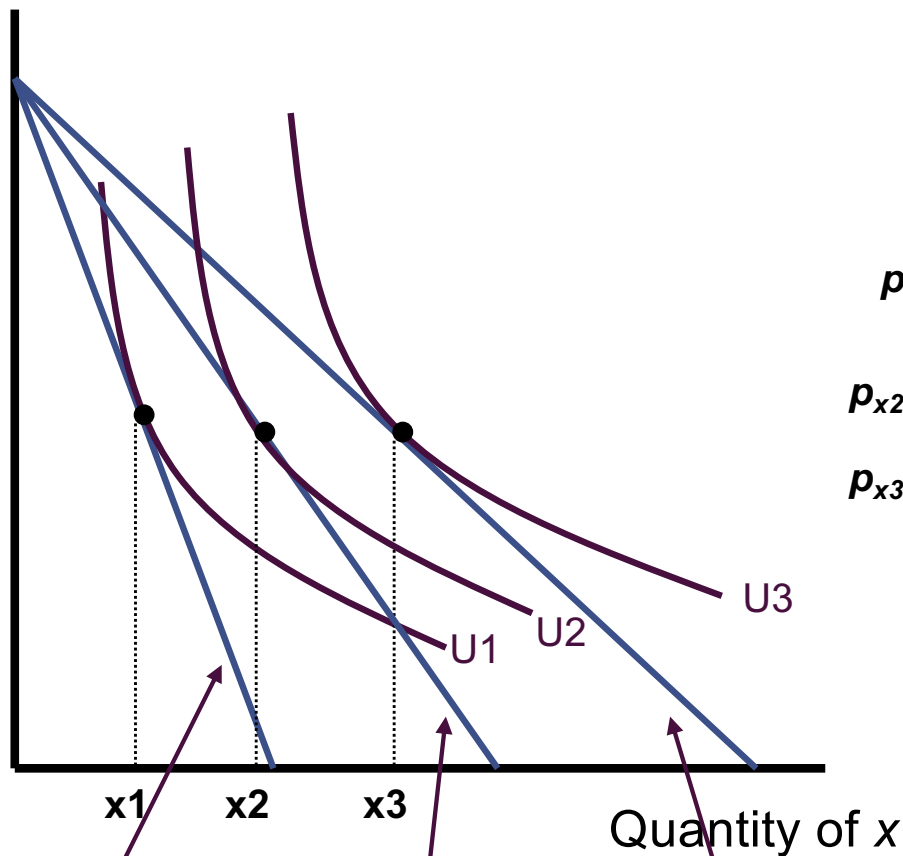
Individual Demand

Consumer Theory

10. Individual Demand

Marshallian demand: Let's see what happens when we vary P_x (ceteris paribus)

Quantity of y

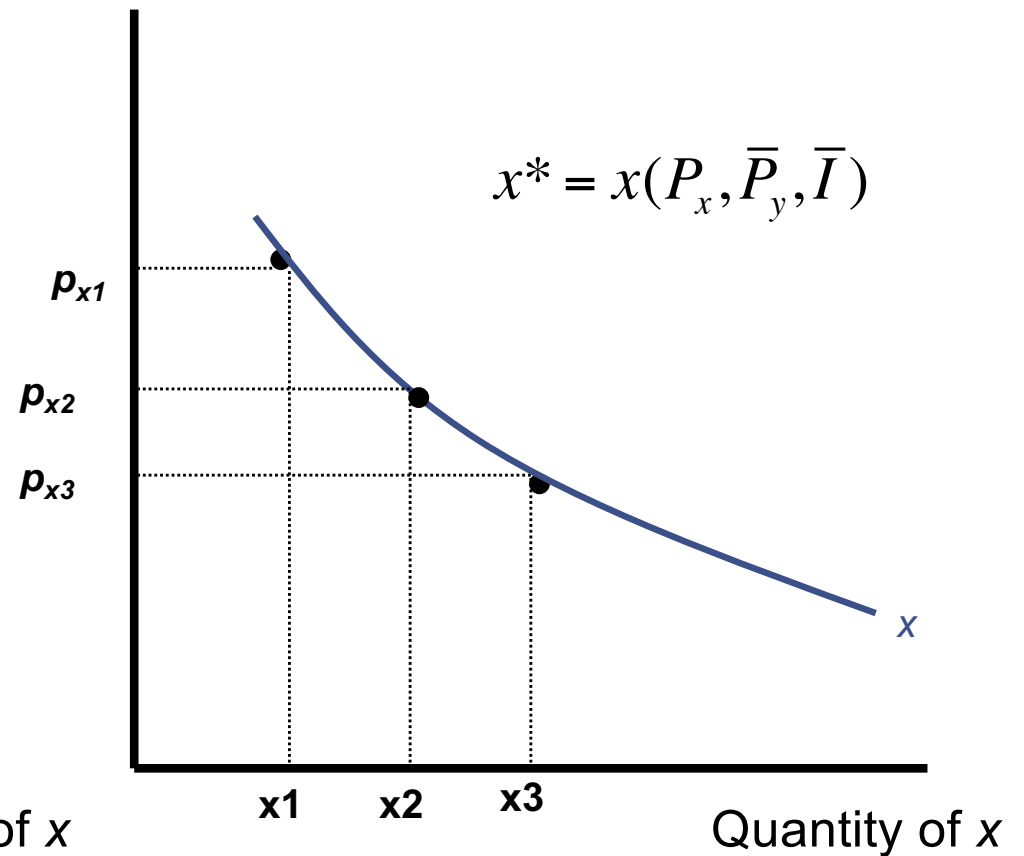


$$I = p_{x1}x + p_y y$$

$$I = p_{x2}x + p_y y$$

$$I = p_{x3}x + p_y y$$

p_x



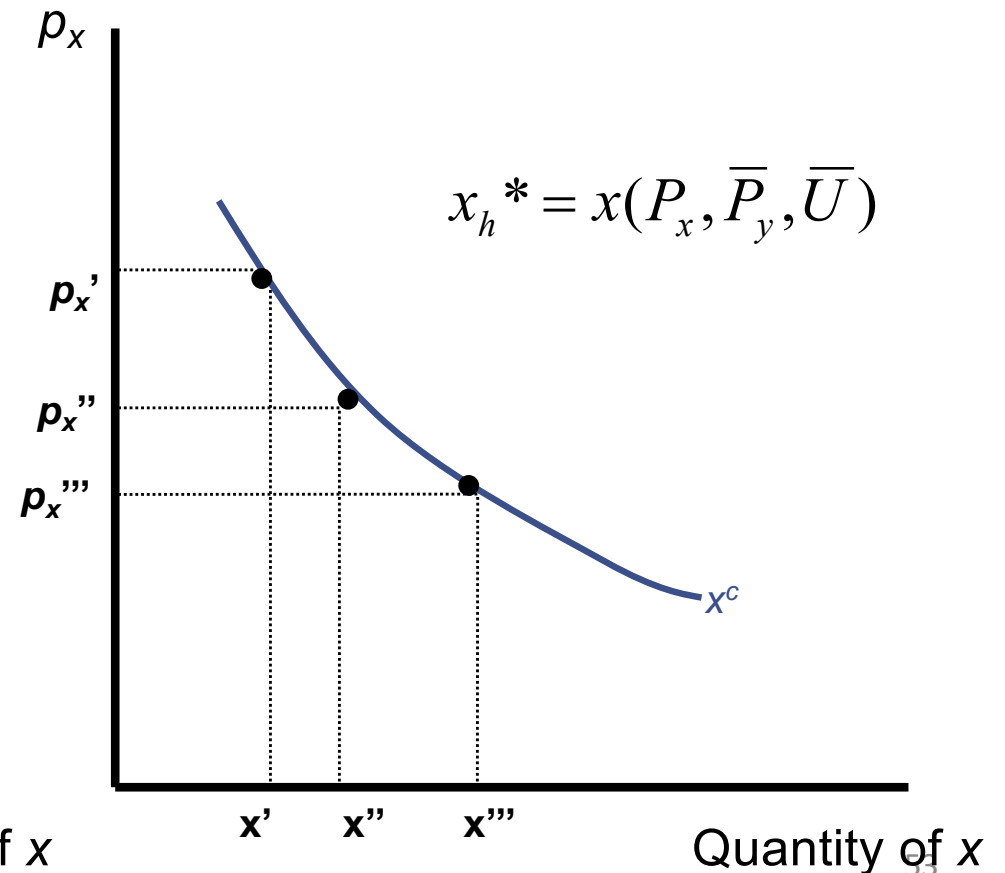
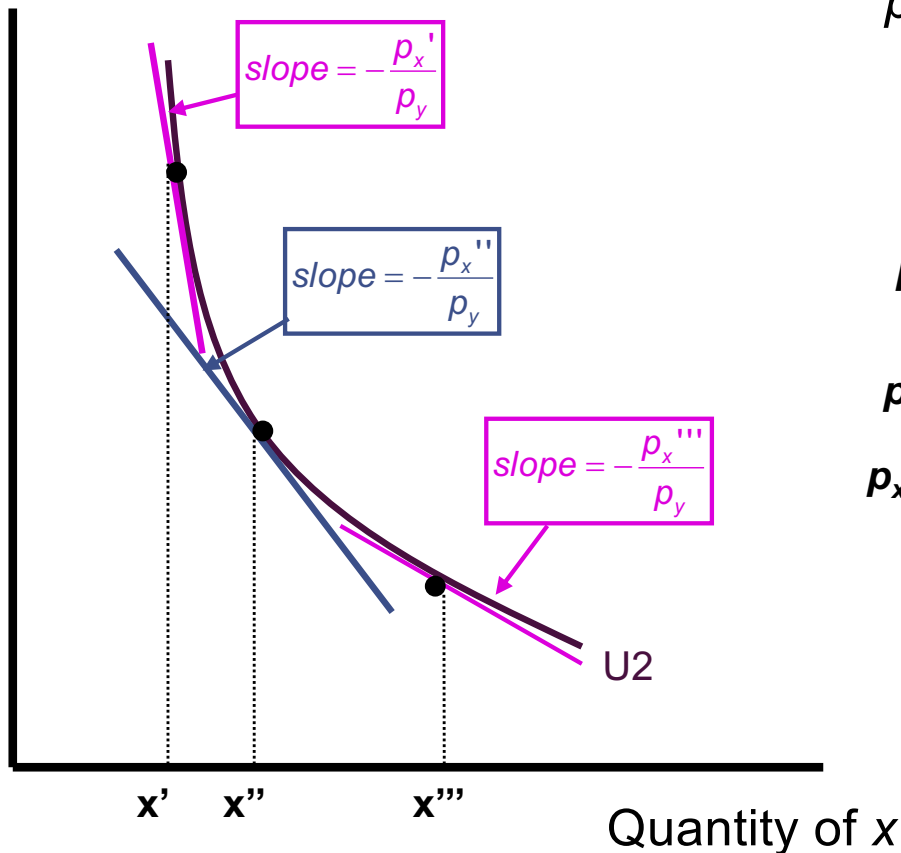
$$x^* = x(P_x, \bar{P}_y, \bar{I})$$

10. Individual Demand

Construction of a **compensated demand (Hicksian) curve**:

Shows the relationship between the price of a good and the quantity purchased on the assumption that **other prices and utility are held constant**

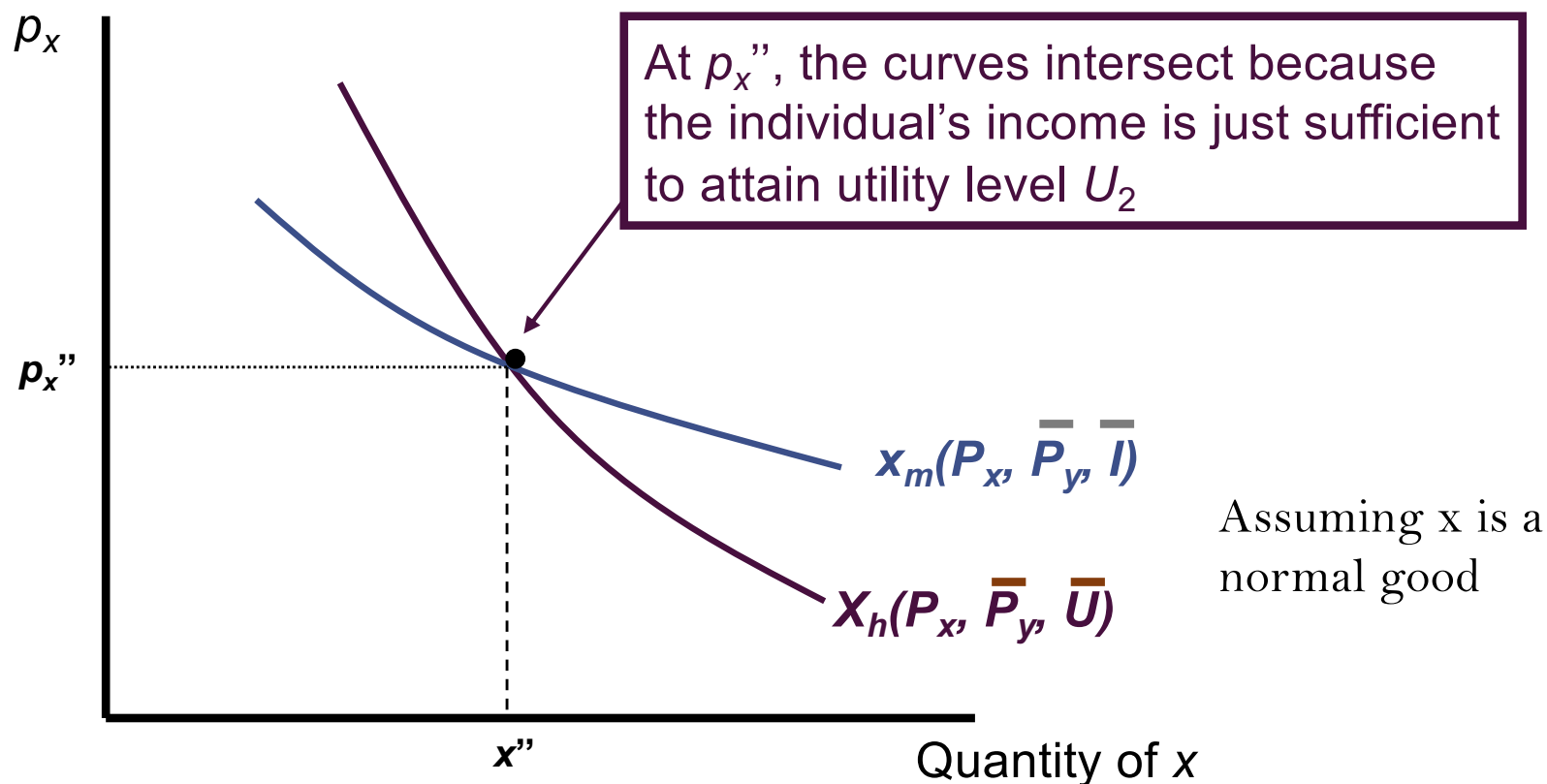
Quantity of y



10. Individual Demand

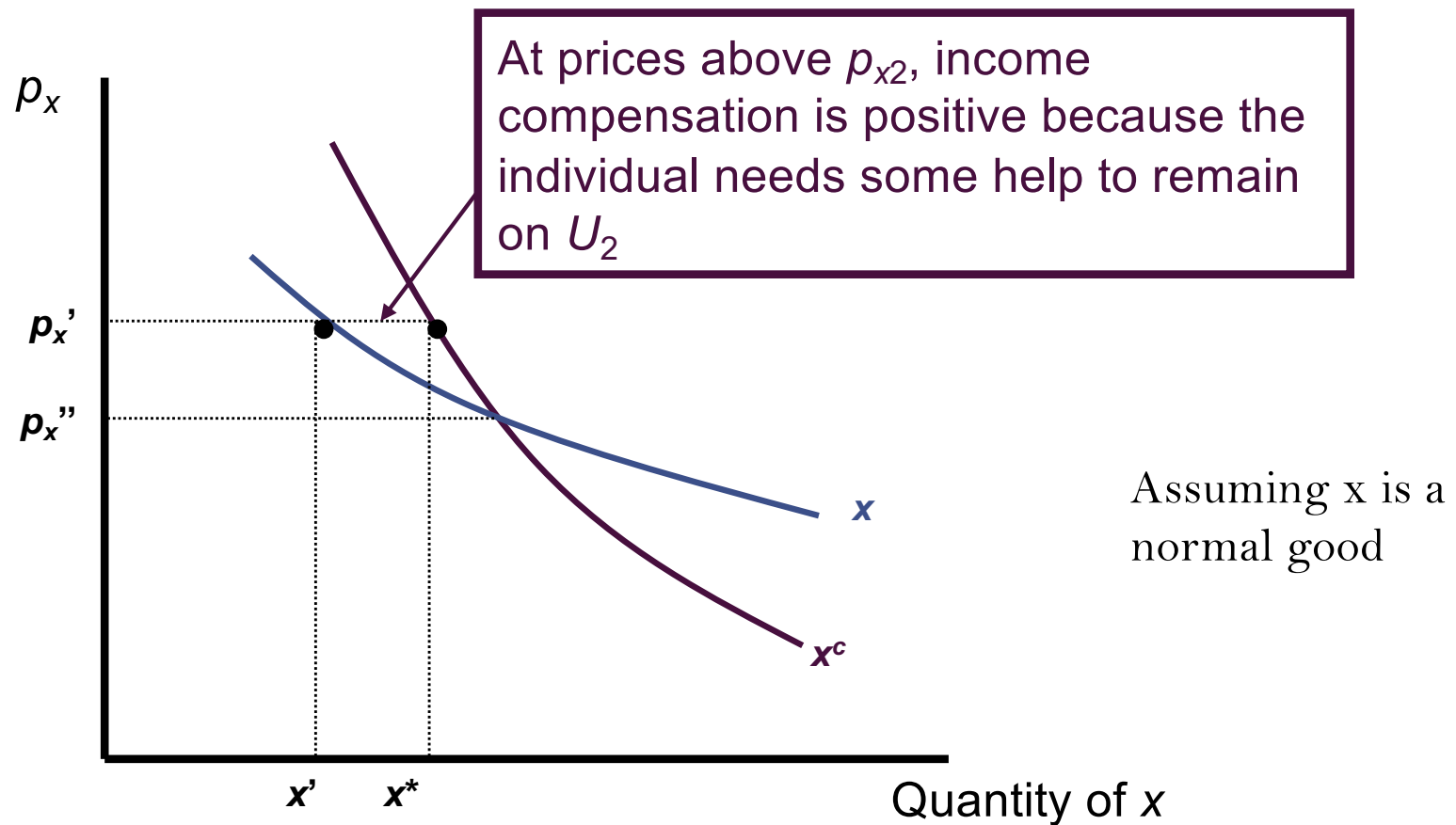
Comparison of Hicksian and Marshallian demand curves:

For a normal good, the compensated demand curve is less responsive to price changes than is the uncompensated demand curve



10. Individual Demand

Comparison of Hicksian and Marshallian demand curves:



10. Individual Demand

Comparison of Hicksian and Marshallian demand curves:

